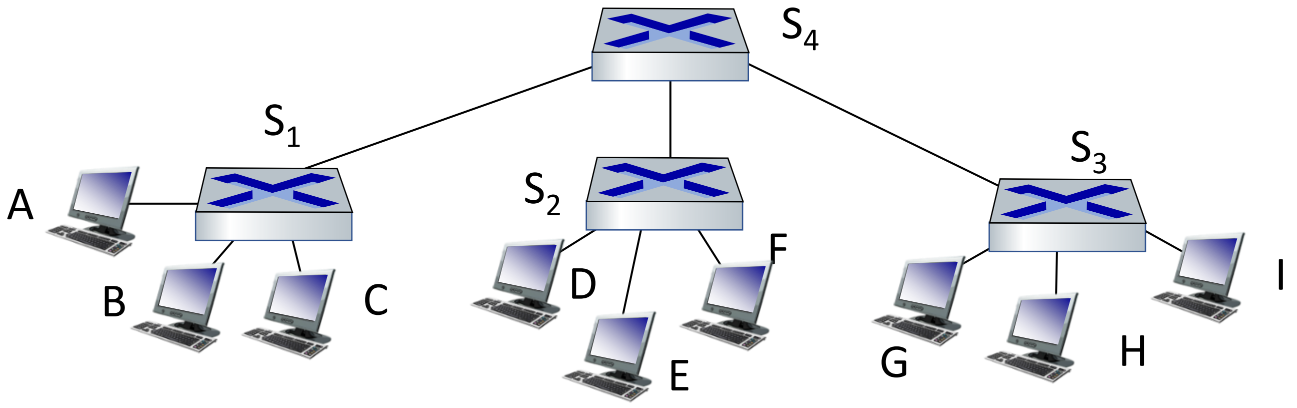
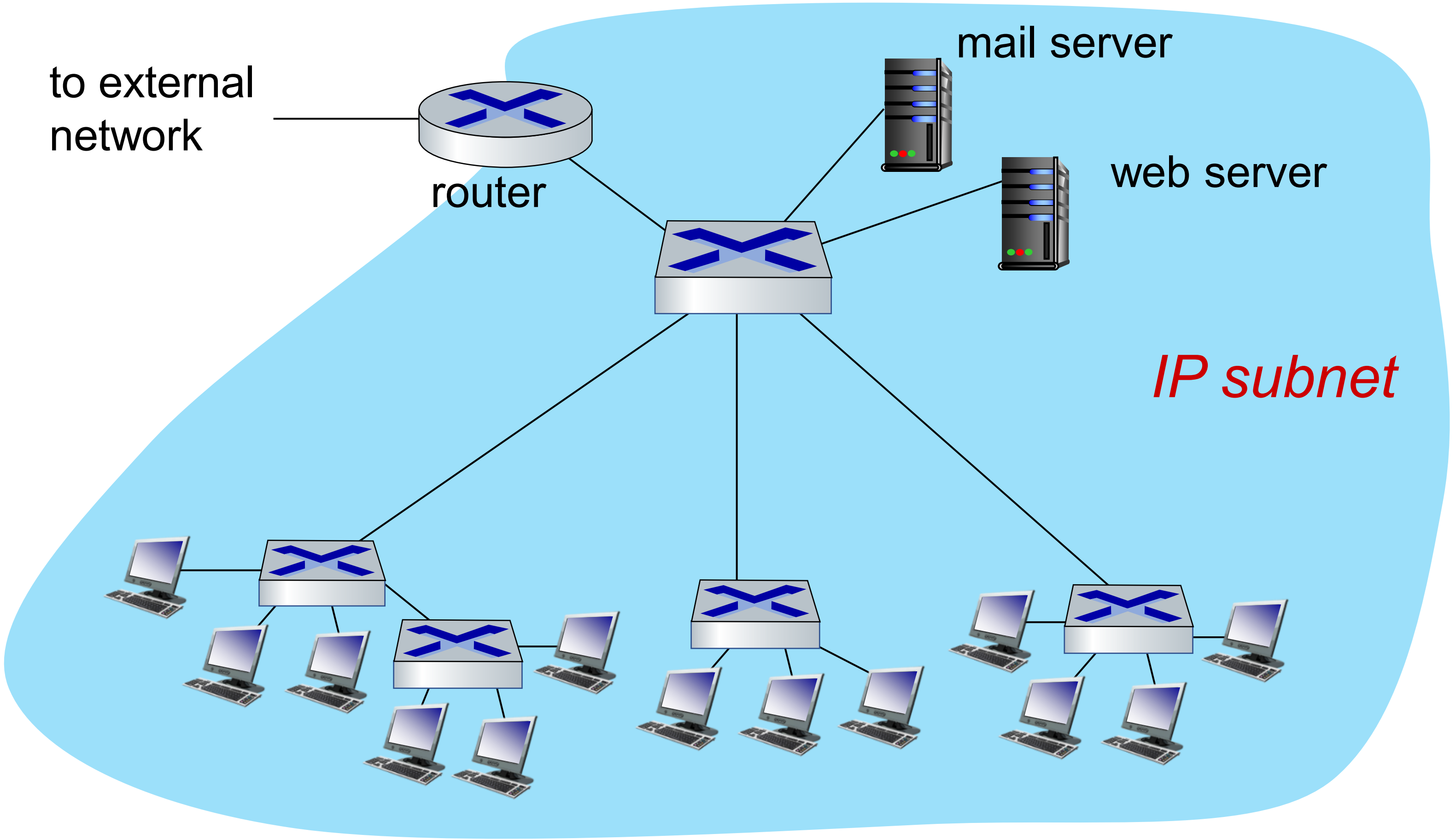
Self-learning multi-switch example

Suppose C sends frame to I, I responds to C



Q: show switch tables and packet forwarding in S_1, S_2, S_3, S_4

Small institutional network



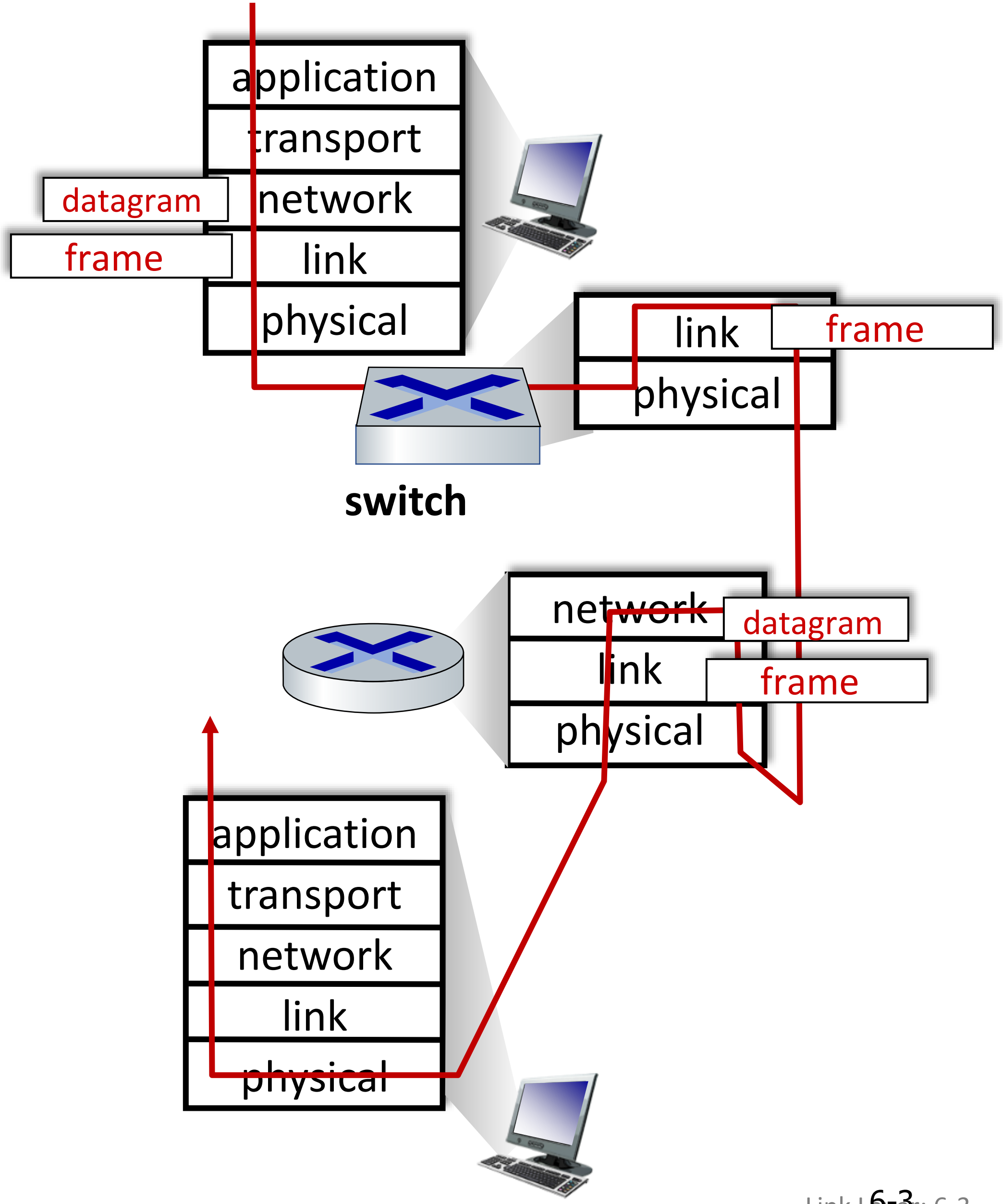
Switches vs. routers

both are store-and-forward:

- *routers*: network-layer devices (examine network-layer headers)
- *switches:* link-layer devices (examine link-layer headers)

both have forwarding tables:

- *routers:* compute tables using routing algorithms, IP addresses
- *switches:* learn forwarding table using flooding, learning, MAC addresses



Link layer, LANs: roadmap

- introduction
- error detection, correction
- multiple access protocols
- LANs
  - addressing, ARP
  - Ethernet
  - switches
  - VLANs
- data center networking

- a day in the life of a web request

Virtual LANs

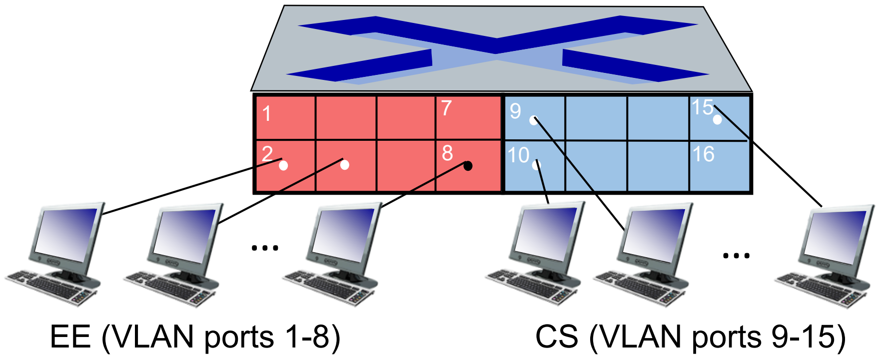
- Divide a single broadcast domain into Multiple Broadcast domains.
- Better performance due to less broadcast message
- A Layer 2 Security (restricted broadcast message)
- Breaking up one physical switch into multiple virtual switches.
- Can be configured on VLAN supported Switches.
- Default VLAN # 1 for all ports if not assigned any VLAN # (2 - 1001)
- Types of VLANs
  - Static VLAN (Based on Port Numbers)
  - Dynamic VLAN (Based on MAC addresses) (Generally we don't use it)

Port-based VLANs

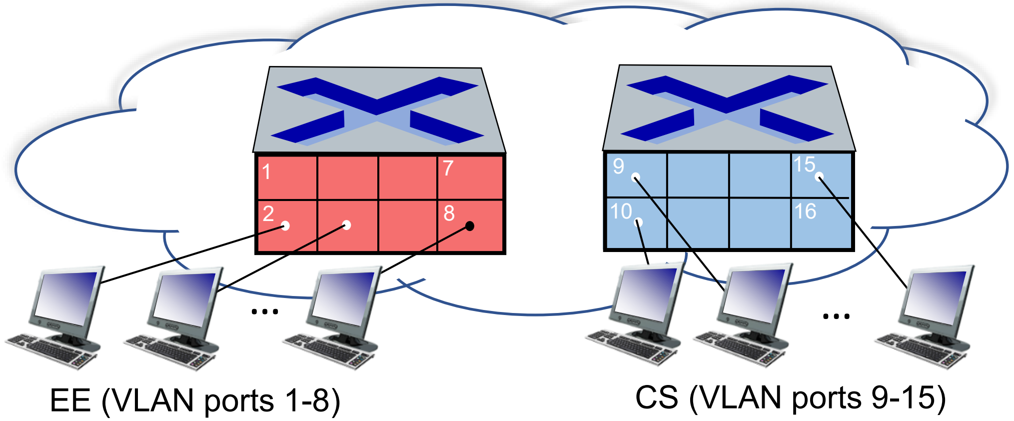
Virtual Local Area Network (VLAN)

switch(es) supporting VLAN capabilities can be configured to define multiple *virtual* LANS over single physical LAN infrastructure.

port-based VLAN: switch ports grouped (by switch management software) so that *single* physical switch ……



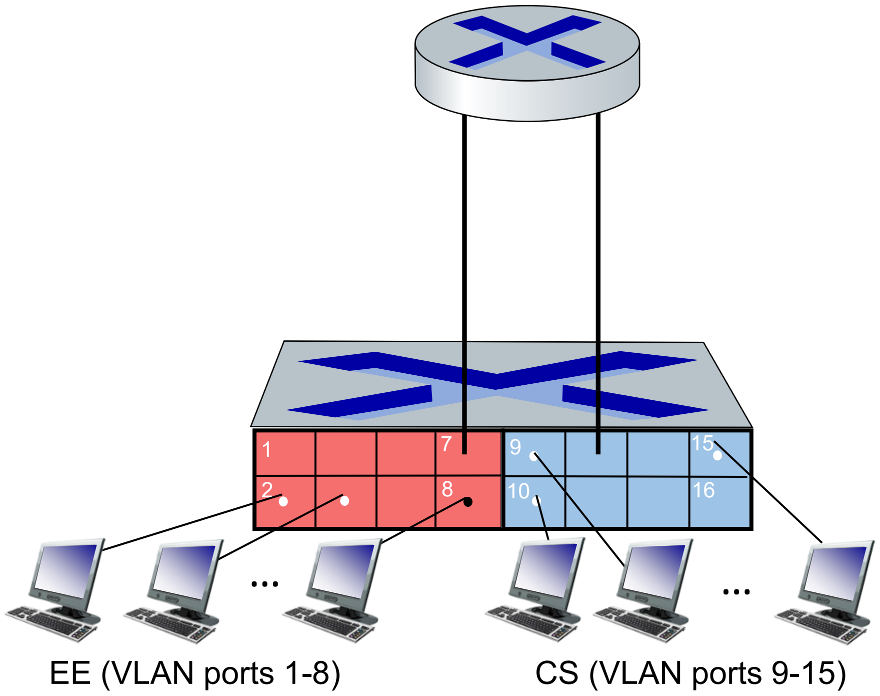
… operates as multiple virtual switches



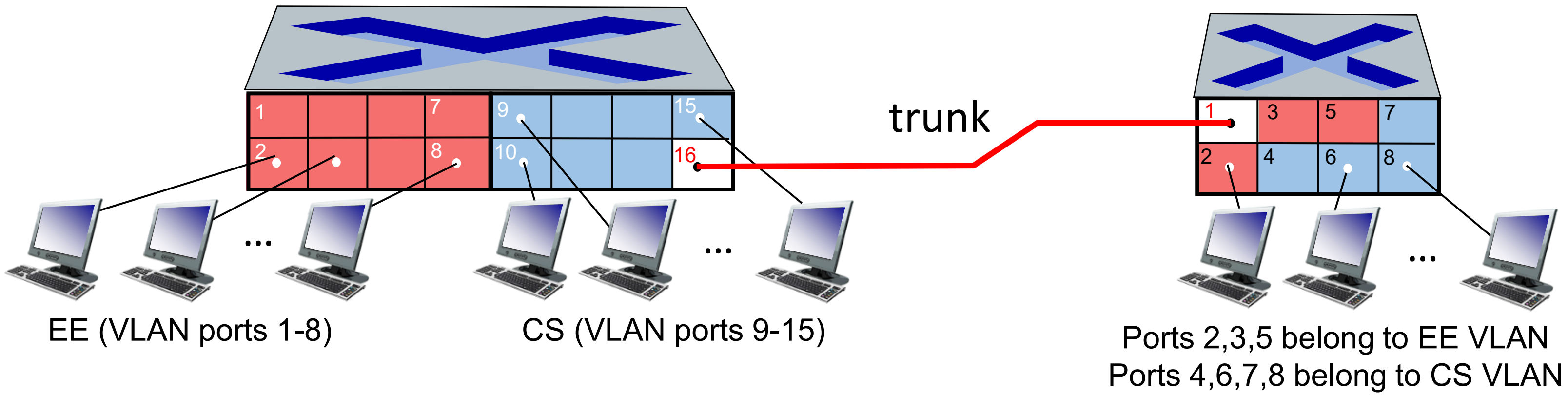
Port-based VLANs

- traffic isolation: frames to/from ports 1-8 can *only* reach ports 1-8
  - can also define VLAN based on MAC addresses of endpoints, rather than switch port

- forwarding between VLANS: done via routing (just as with separate switches)
  - in practice vendors sell combined switches plus routers



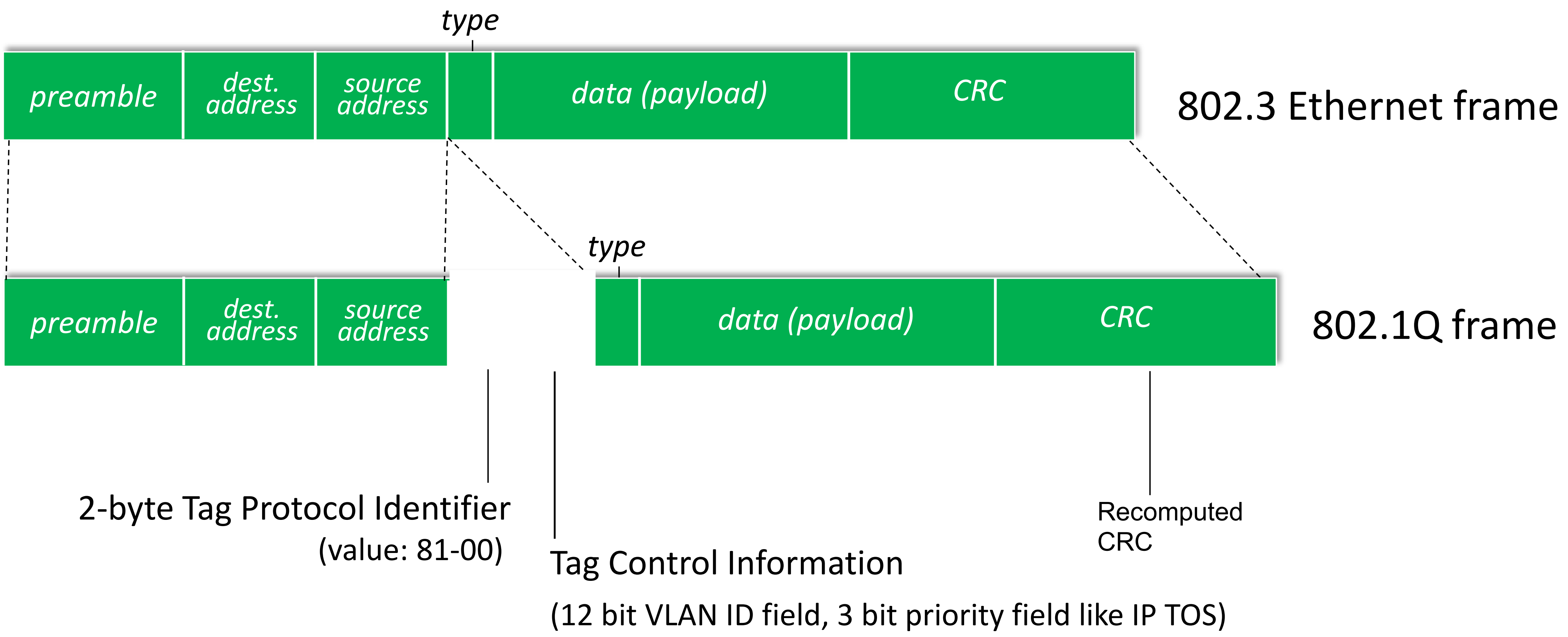
VLANS over multiple switches



Trunk (tagged) port: carries frames between VLANS defined over multiple physical switches

- frames forwarded within VLAN between switches can't be 802.3 frames (must carry VLAN ID info)
- 802.1q protocol adds/removed additional header fields for frames forwarded between trunk ports [Frame tagging]

802.1Q VLAN frame format



802.1Q Frame

```
> Frame 1: 114 bytes on wire (912 bits), 114 bytes captured (912
v Ethernet II, Src: c2:01:2d:ac:00:00 (c2:01:2d:ac:00:00), Dst:
  > Destination: c2:02:18:d4:00:00 (c2:02:18:d4:00:00)
  > Source: c2:01:2d:ac:00:00 (c2:01:2d:ac:00:00)
    Type: IPv4 (0x0800)
> Internet Protocol Version 4, Src: 10.12.10.1, Dst: 10.23.20.3
> Internet Control Message Protocol
```

```
> Frame 2: 118 bytes on wire (944 bits), 118 bytes captured (944
v Ethernet II, Src: c2:01:2d:ac:00:00 (c2:01:2d:ac:00:00), Dst:
  > Destination: c2:02:18:d4:00:00 (c2:02:18:d4:00:00)
  > Source: c2:01:2d:ac:00:00 (c2:01:2d:ac:00:00)
    Type: 802.1Q Virtual LAN (0x8100)
v 802.1Q Virtual LAN, PRI: 0, CFI: 0, ID: 10
    000. .... .... .... = Priority: Best Effort (default) (0)
    ...0 .... .... .... = CFI: Canonical (0)
    .... 0000 0000 1010 = ID: 10
    Type: IPv4 (0x0800)
> Internet Protocol Version 4, Src: 10.12.10.1, Dst: 10.23.20.3
> Internet Control Message Protocol
```

Fig Source: https://www.practicalnetworking.net/stand-alone/vlans/